

SIGHETUL MARMATIEI, Romania

WMO: 150040

Lat: 47.9392N

Lon: 23.9044E

Elev: 906

StdP: 14.22

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	9.9	-0.5	5.5	6.5	4.3	7.1	12.1	12.2	32.3	9.8	31.1	2.4	140	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	88.5	69.3	85.6	68.0	82.7	66.5	71.4	84.2	69.7	81.9	68.2	79.6	3.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	103.9	76.2	65.6	97.7	74.5	64.1	92.7	73.2	35.7	84.6	34.2	82.0	32.9	79.8	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.2	8.9	7.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-3.1	92.7	7.9	2.2	-8.7	94.3	-13.4	95.6	-17.8	96.9	-23.5	98.5
			-3.4	74.8	7.8	1.7	-9.0	76.0	-13.5	77.0	-17.9	78.0	-23.5	79.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	28.5	33.4	40.2	51.4	59.6	65.9	68.6	68.7	59.7	50.8	40.9	29.7
	DBStd	16.08	8.39	8.01	7.39	7.52	6.00	5.78	5.28	5.53	6.97	7.48	8.15	9.36
	HDD50	2531	666	464	310	73	5	0	0	0	6	86	291	630
	HDD65	5935	1131	884	769	407	185	58	27	25	188	442	724	1095
	CDD50	2478	0	0	6	116	301	477	577	579	296	109	17	0
	CDD65	411	0	0	0	1	16	86	139	140	28	1	0	0
	CDH74	4142	0	0	0	42	252	858	1299	1360	315	16	0	0
	CDH80	1278	0	0	0	3	37	253	417	488	80	0	0	0
Wind	WSAvg	2.5	1.9	2.3	3.2	3.1	2.8	2.6	2.6	2.4	2.3	2.3	2.3	1.7
Precipitation	PrecAvg	33.8	2.0	1.9	2.2	2.3	3.5	3.9	4.1	3.1	3.1	2.8	2.4	2.4
	PrecMax	43.9	4.8	5.7	6.0	3.6	7.8	6.9	9.0	9.3	6.1	6.6	5.3	4.2
	PrecMin	25.7	0.9	0.7	0.4	0.8	1.0	1.4	0.8	0.7	0.7	0.3	0.2	0.4
	PrecStd	5.5	0.9	1.0	1.5	0.8	1.4	1.6	2.3	1.7	1.4	1.7	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	57.5	66.2	78.4	83.6	89.3	91.2	93.0	87.4	76.1	66.3	50.4
		MCWB	43.8	47.4	51.0	58.1	63.6	70.2	70.2	69.5	66.7	60.7	54.8	45.3
	2%	DB	44.5	51.1	59.5	73.9	80.3	86.2	87.7	89.0	81.4	71.8	60.9	46.4
		MCWB	41.2	44.2	48.1	56.9	62.5	68.1	69.6	69.0	63.6	59.1	52.4	43.0
	5%	DB	41.3	46.6	55.3	70.0	76.6	82.9	84.7	85.4	77.5	67.1	56.6	43.2
		MCWB	38.7	42.1	45.9	54.7	61.2	66.6	68.1	68.3	62.8	57.3	50.0	41.2
	10%	DB	38.4	43.1	51.3	65.9	73.2	79.0	81.5	81.7	72.7	62.0	52.3	40.7
		MCWB	36.3	39.4	44.1	52.6	59.7	64.9	67.3	66.7	61.1	55.2	48.0	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	48.6	53.1	60.3	66.6	73.5	72.8	73.9	68.1	62.3	56.1	46.6
		MCDB	47.4	57.2	62.4	74.1	79.1	86.0	85.8	86.9	83.3	72.0	63.3	50.3
	2%	WB	41.6	45.0	49.4	58.0	64.4	69.9	71.1	71.5	66.1	60.3	53.5	43.4
		MCDB	44.1	50.1	58.0	71.0	76.4	82.7	84.3	84.0	78.3	69.4	59.0	45.7
	5%	WB	38.9	42.6	47.2	56.0	62.8	68.1	69.7	69.4	64.1	58.3	51.1	41.5
		MCDB	40.9	46.3	53.8	67.6	73.8	79.6	81.7	81.2	74.3	65.5	56.2	43.3
	10%	WB	36.7	39.9	44.8	54.0	60.8	66.5	68.1	67.8	62.1	55.8	48.3	39.4
		MCDB	38.2	42.5	50.2	63.8	70.1	76.7	78.8	79.6	70.3	61.3	51.9	40.9
Mean Daily Temperature Range	5% DB	MDBR	11.5	14.2	17.6	21.2	21.7	21.3	21.2	22.9	19.9	17.9	13.6	11.0
		MCDBR	12.7	18.4	23.8	28.2	27.9	26.8	26.7	27.8	27.5	25.3	21.0	12.2
		MCWBR	8.9	11.7	13.6	13.1	12.3	11.1	10.7	10.3	11.5	13.5	12.5	9.0
	5% WB	MCDBR	11.3	16.8	21.0	25.7	24.7	24.2	24.0	24.8	23.9	22.6	18.3	11.6
		MCWBR	8.3	11.0	12.7	12.5	11.4	11.0	10.5	10.7	10.8	12.3	11.3	8.7
Clear-Sky Solar Irradiance	taub	0.321	0.351	0.401	0.459	0.441	0.437	0.461	0.461	0.418	0.384	0.353	0.322	
	taud	2.230	2.157	2.107	2.043	2.166	2.217	2.161	2.154	2.231	2.293	2.308	2.293	
	Ebn at Noon	231	248	254	251	262	263	254	248	247	236	218	215	
	Edh at Noon	25	34	42	49	45	43	45	44	37	30	24	21	
All-Sky Solar Radiation	RadAvg	332	590	954	1358	1663	1790	1761	1620	1136	732	396	249	
	RadStd	52	82	135	142	174	129	162	150	153	102	56	37	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (44 neighbors)	+1.26	N/A	N/A	+1.87	N/A	N/A	-270	-389	+189	+70	

Nomenclature: See separate page